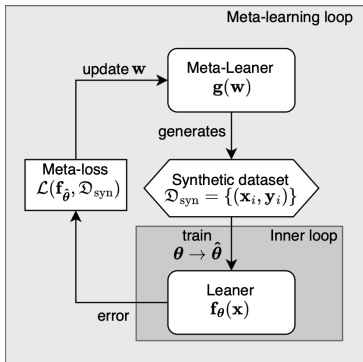


# Extracting inductive bias via generating easy-to-generalize datasets



- ▶  $\mathcal{D}_{\text{syn}} = \{(\mathbf{x}_i, \mathbf{y}_i)\} = \mathcal{D}_{\text{train}} \cup \mathcal{D}_{\text{test}}$  is produced by generator  $g(\mathbf{w})$
- ▶  $f_{\hat{\theta}}$  is the target model, trained on  $\mathcal{D}_{\text{train}}$
- ▶  $\mathcal{L} = \text{error} + \mathcal{R}$  is the meta-loss function:
  - Error can be Cross-entropy or MSE on  $\mathcal{D}_{\text{test}}$
  - $\mathcal{R}$  is a regularization term

We meta-train the generator  $g(\mathbf{w})$  to find parameters  $\mathbf{w}^*$  that minimize the meta-loss:

$$\mathbf{w}^* = \arg \min_{\mathbf{w} \in \mathbb{W}} E_{\mathcal{D}_{\text{syn}} \sim g(\mathbf{w})} [\mathcal{L}(f_{\hat{\theta}}, \mathcal{D}_{\text{syn}})]$$